Module Project On -

Dimensionality Reduction

Presented By –

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Problem Statement - Student Performance Clustering

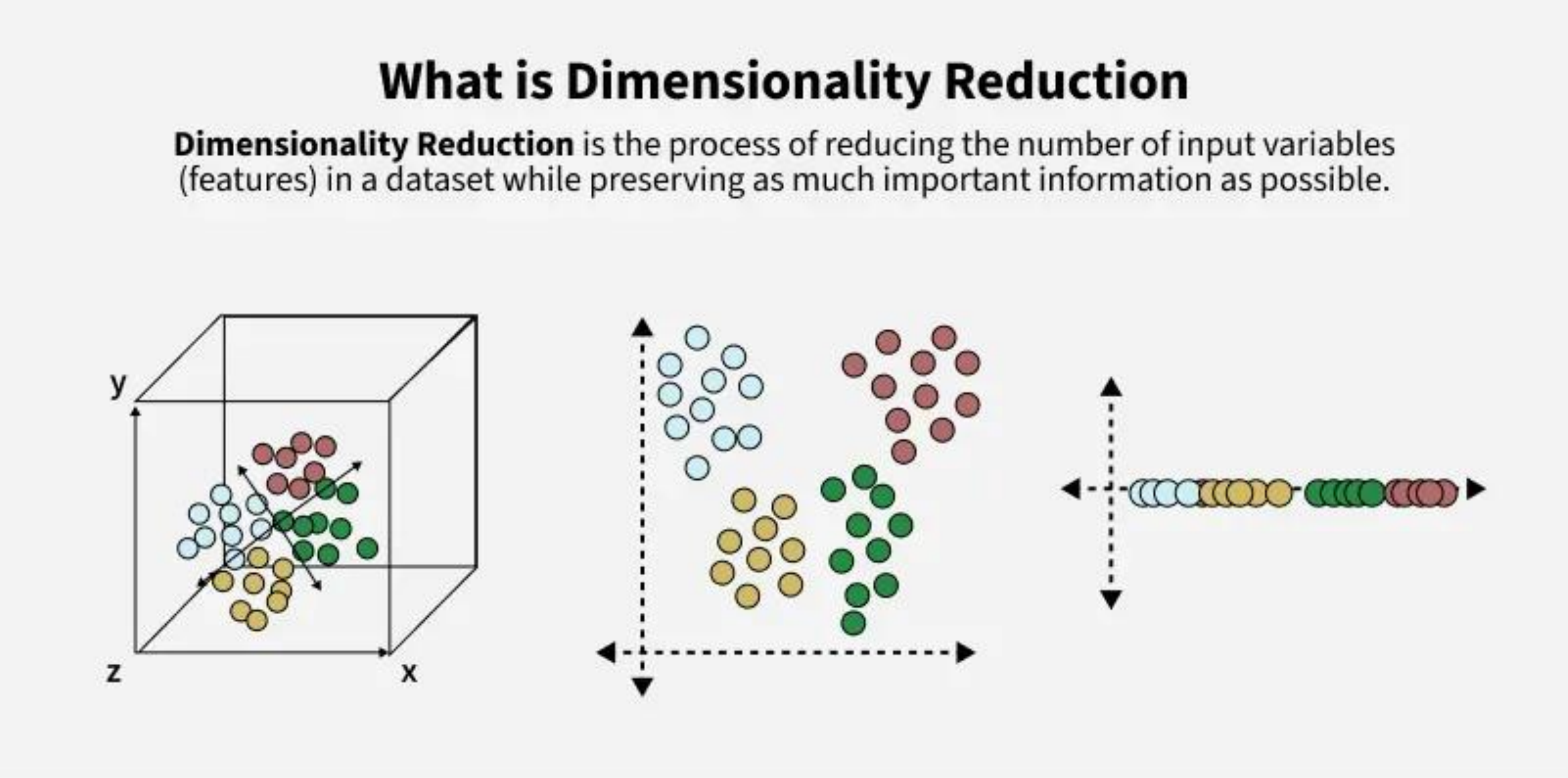
Goal

- The primary goal of this project is to use **dimensionality reduction techniques** to analyze high-dimensional student performance data, aiming to uncover natural groupings or segments among the students. These segments can reveal hidden patterns of performance (e.g., students who excel in STEM but struggle in humanities, or vice-versa) that are not immediately obvious from raw grade averages.

Objectives

- **Dimensionality Reduction:** Apply **Principal Component Analysis (PCA)** to reduce the feature space (grades across all subjects) to 2 or 3 principal components.
- **Visualization Comparison:**
  - Plot the student data using the reduced PCA components.
  - Apply a **non-linear technique** like **t-SNE** or **UMAP** to the data and plot the results.
- **Analysis & Interpretation:** Visually compare the outputs of PCA and t-SNE/UMAP to **identify clusters** of students. Interpret the meaning of these clusters (e.g., "Cluster A" represents students who score consistently high across all subjects).

Dimensionality Reduction ?



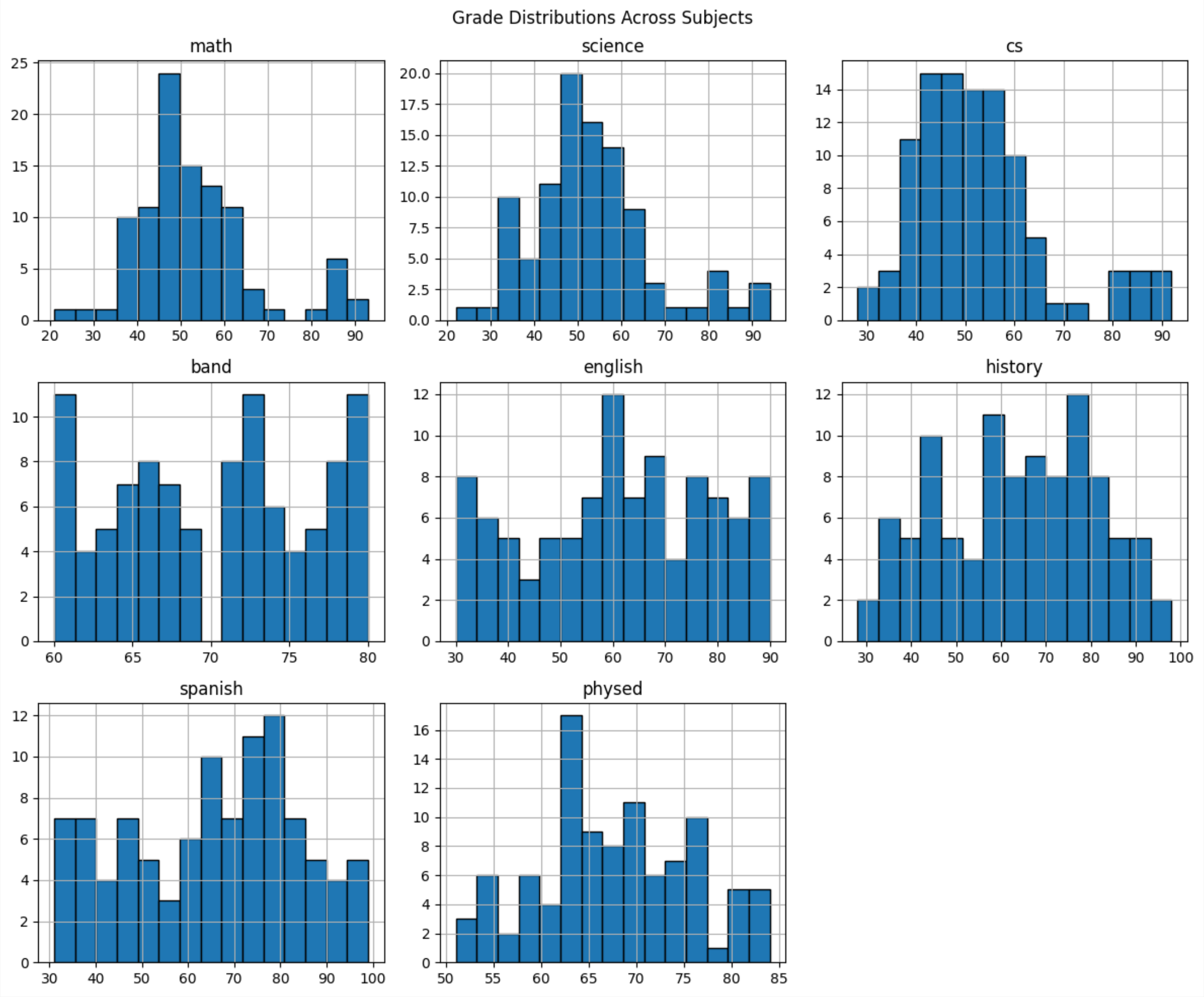
Dataset

Student Grades Data: The dataset contains individual student grades across multiple subjects, including: **Math**, **Science**, **Computer Science (CS)**, **Band**, **English**, **History**, **Spanish**, and **Physical Education (PhysEd)**, along with a unique student_id.

	student_id	math	science	cs	band	english	history	spanish	physed
0	1	46	48	50	74	34	44	39	73
1	2	66	65	65	66	74	80	75	63
2	3	55	53	50	76	71	72	76	71
3	4	53	57	53	80	77	77	85	82
4	5	55	62	58	67	82	77	78	60

Column Name	Description
student_id	Unique identifier for each student.
math	Grade in Mathematics.
science	Grade in Science.
cs	Grade in Computer Science.
band	Grade in Band (or a similar elective).
english	Grade in English.
history	Grade in History.
spanish	Grade in Spanish (or a foreign language).
physed	Grade in Physical Education.

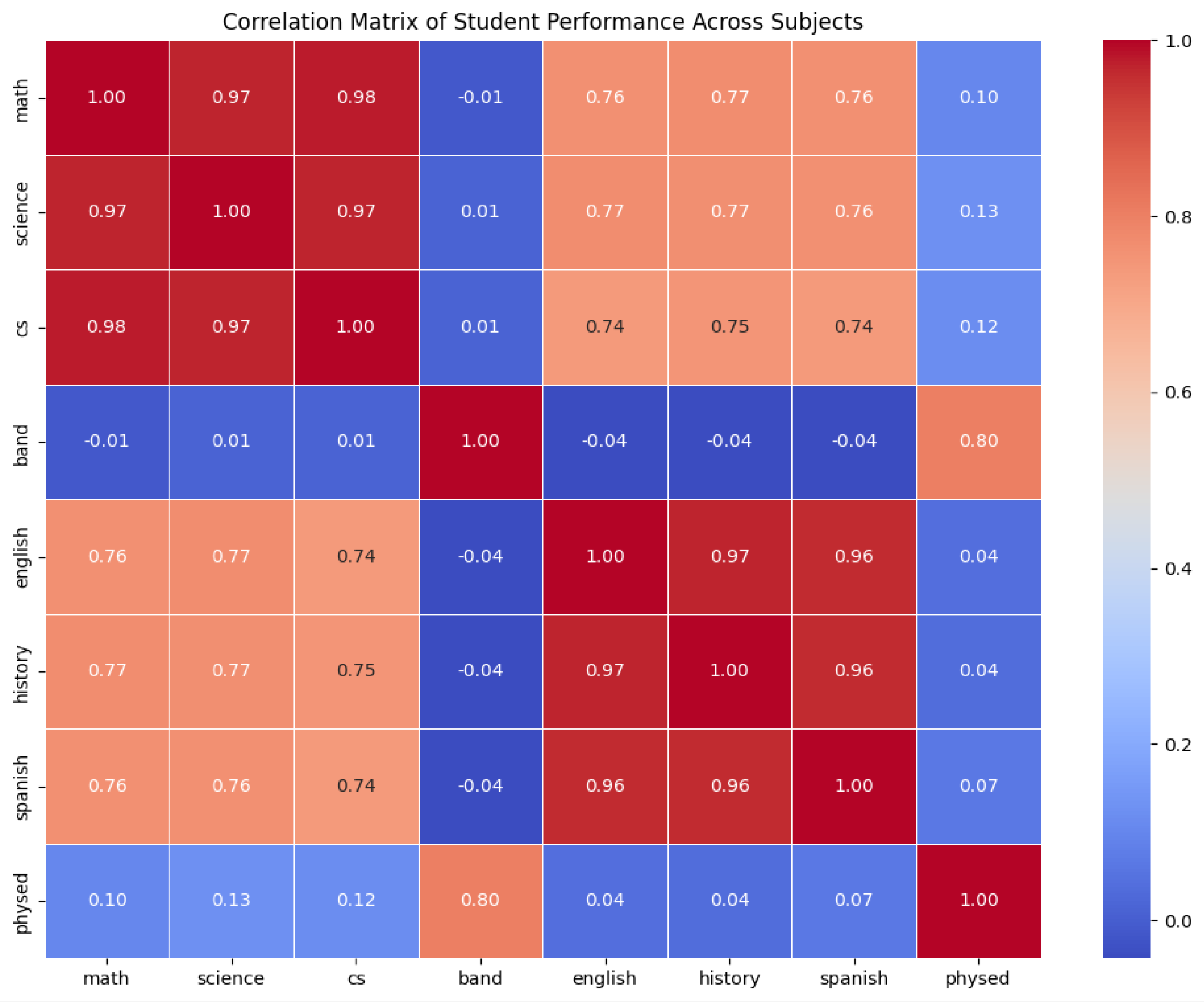
EDA and Feature Engineering



Math, Science, & CS (STEM): Follow tight, bell-shaped curves centered around a ~50 average, featuring small, isolated groups of high-performing outliers (85+).

Band, PhysEd, & Humanities: Show highly spread out, multi-peaked distributions; Band and PhysEd have strict grade floors (nothing below 60 and 50 respectively), while English, History, and Spanish span widely from 30 to 100.

EDA and Feature Engineering



•**Two Major Academic Blocks:** There are two massive clusters. Math, Science, and Computer Science are almost perfectly linked together (around 0.97 correlation). English, History, and Spanish form a second tight group (around 0.96 correlation).

•**Strong Academic Cross-Over:** Students who do well in STEM also tend to do very well in Humanities, showing a strong cross-discipline connection (around 0.75 correlation) across all traditional subjects.

•**The Creative/Physical Exception:** Band and Physical Education have virtually zero connection to core academic grades. Instead, they link heavily with each other (0.80 correlation), defining a completely separate group of students.

EDA and Feature Engineering

FEATURE ENGINEERING DETAILS

*** Feature Name: STEM_Avg**

- Formula: (math + science + cs) / 3
- Operational Purpose: **Aggregates raw performance across core quantitative and technical subjects into a single robust baseline.**
- **Role in PCA Loading: Holds a heavy positive loading on PC1 (0.346), strongly driving the general academic competence vector while maintaining minimal weight on PC2.**

*** Feature Name: Humanities_Avg**

- Formula: (english + history + spanish) / 3
- Operational Purpose: **Aggregates performance across linguistic, reading-heavy, and contextual domains to evaluate text-comprehension trends.**
- Role in PCA Loading: Holds a heavy positive loading on PC1 (0.359), aligning tightly with the general performance axis while carrying a slight negative weight on PC2.

*** Feature Name: Electives_Avg**

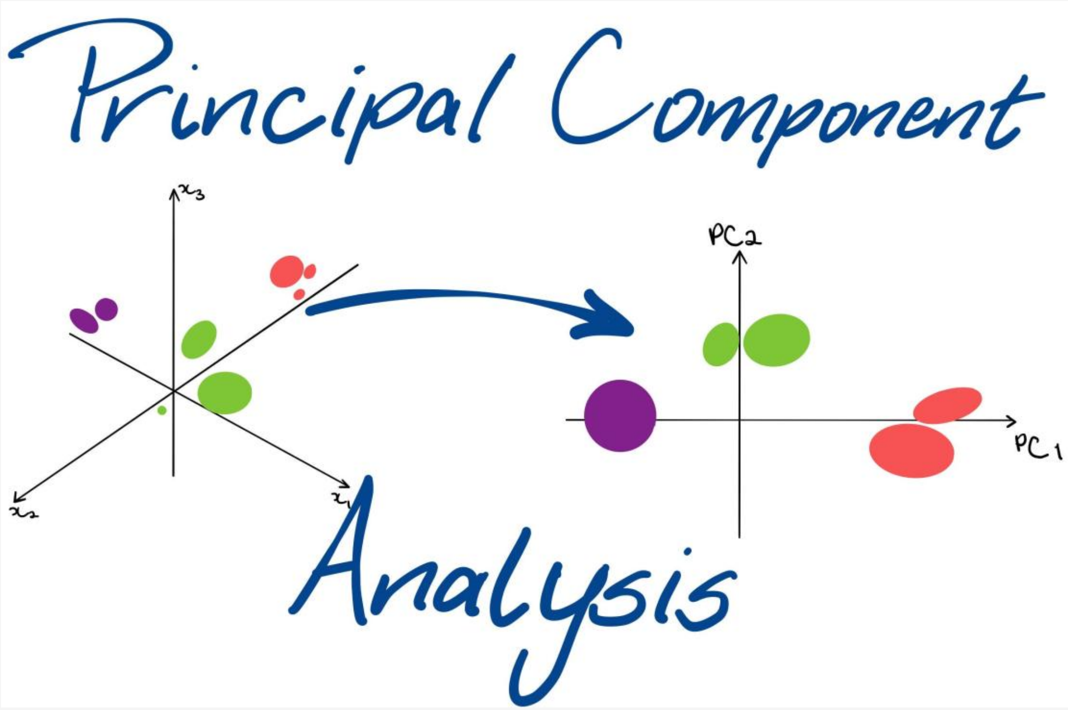
- Formula: (band + physed) / 2
- Operational Purpose: **Tracks student performance in performance-based, active, and highly practical domains outside the traditional classroom environment.**
- Role in PCA Loading: Holds a dominant positive loading on PC2 (0.586), serving as the primary mathematical anchor for the secondary Elective Specialization axis.

*** Feature Name: STEM_Humanities_Gap**

- Formula: STEM_Avg - Humanities_Avg
- Operational Purpose: **Measures the explicit, directional performance bias of a student to see whether they lean toward analytical or linguistic subjects.**
- Role in PCA Loading: Exhibits a bipolar loading profile (-0.141 on PC1, 0.136 on PC2), acting as a key contrast indicator between the two core academic tracks.

PCA (Principal Component Analysis)

Finds the directions (principal components) that maximize the variance in the data.



Step-by-Step Implementation

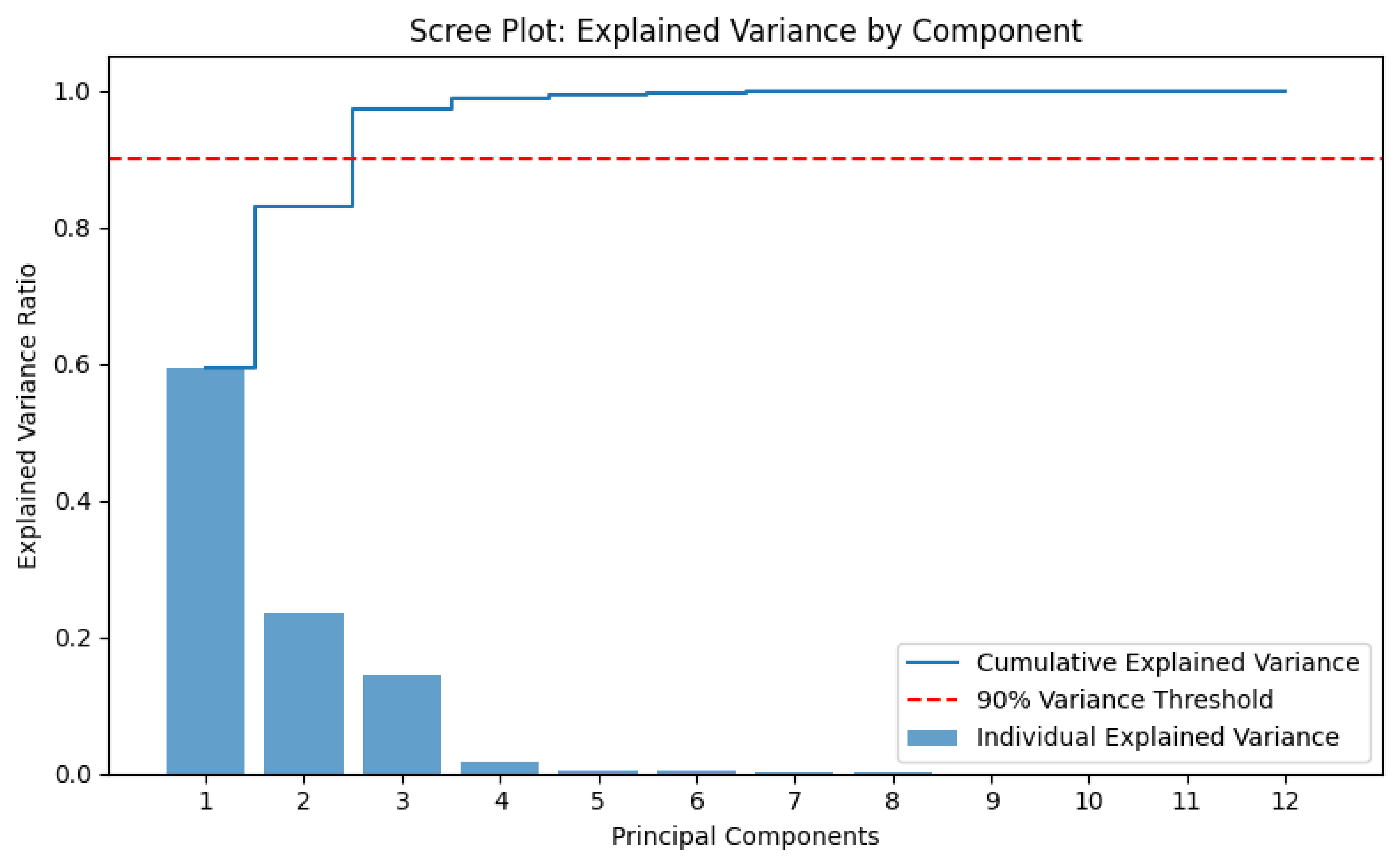
Standardize the data: Because PCA is highly sensitive to the scale of features, you must scale your data so that each feature has a mean of 0 and a standard deviation of 1.

Initialize PCA: Choose the number of components or the percentage of variance you want to retain.

Fit and Transform: Pass the standardized data to the algorithm to generate your new principal components.

PCA (Principal Component Analysis)

Explained Variance By Components



Based on the Scree Plot visualization, the optimal number of principal components to choose is **3**.

•**Variance Threshold Optimization:** The orange line tracks the cumulative explained variance. Component 3 successfully crosses the target **85% Cumulative Variance Threshold** (represented by the red dashed line).

PCA (Principal Component Analysis)

Reduce feature space to 2 components for 2D visual comparison

Principal Component 1 (PC1): Overall Academic Aptitude [0.344, 0.344, 0.338, -0.002, 0.354, 0.355, 0.353, 0.038, 0.345, 0.359, 0.021, -0.141]
Interpretation: PC1 represents overall academic competence and student drive.
Reasoning: Notice that almost all core subjects (indices 0–2, 4–6, and 8–9) share nearly identical, high positive weights ranging tightly between 0.33 and 0.36. When features move together in the same positive direction with equal intensity, the component captures the collective baseline performance. A high positive score on PC1 simply means the student gets high grades across the board, regardless of the domain.
Principal Component 2 (PC2): The Elective vs. Core Academic Tradeoff [0.033, 0.047, 0.049, 0.553, -0.054, -0.054, -0.045, 0.559, 0.043, -0.051, 0.586, 0.135]
Interpretation: PC2 captures a strong non-academic elective specialization profile (specifically separating performance in Band, Physical Education, and your third engineered elective feature from core subjects).
Reasoning: Look at the massive positive weights at index 3 (0.553), index 7 (0.559), and index 10 (0.586). These point directly to your non-core elective categories like band and physed and their engineered averages. Meanwhile, the core academic subjects drop down close to zero or flip slightly negative (-0.05).
A high positive score on PC2 isolates a specific cluster of students: those who excel heavily in performance, arts, and physical tracking over traditional classroom settings.

PCA (Principal Component Analysis)

Explained Variance Ratio

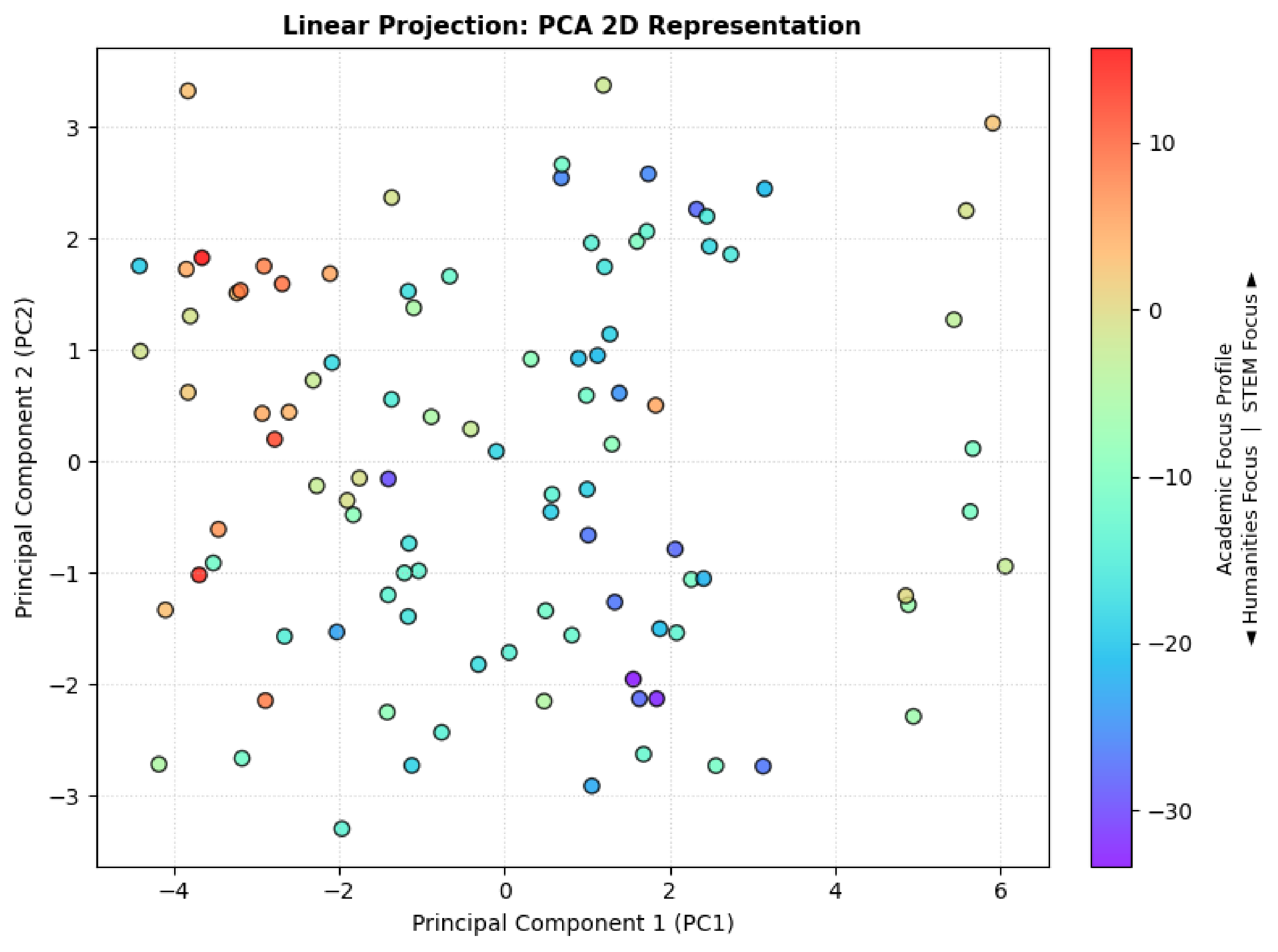
Principal Component 1 (PC1) Variance Capture Value: 0.59295083 (or 59.30%). Interpretation: **The first principal component alone captures nearly 60% of the total variance present in your entire 12-feature student dataset**. Because it accounts for the clear majority of the data's spread, it represents the primary underlying factor driving student performance variations (typically overall academic aptitude and grade scaling).

Principal Component 2 (PC2) Variance Capture Value: 0.23631401 (or 23.63%). Interpretation: **The second principal component accounts for nearly a quarter of the remaining variance in the data.** This component isolates the secondary major trend in your student population, which—as seen in your loadings—captures the distinct performance tradeoff between core academic subjects and non-academic electives (like Band and PhysEd).

Cumulative 2D Variance Significance Combined Value: 0.5930+0.2363=0.8293 (or 82.93%). Interpretation: Together, compressing your 12 continuous grade features down to just 2 principal components retains roughly 83% of the original information. This confirms that a 2D projection is highly efficient for your dataset; it eliminates 10 dimensions of data density while losing only ~17% of the total variance, making it completely viable for distinct visual clustering and student profiling.

PCA (Principal Component Analysis)

Visualize the PCA components




```
--- Latent Factor Matrix: Principal Component Loadings ---
                           PC1       PC2
math                  0.344092  0.033322
science               0.344166  0.047628
cs                    0.338901  0.049058
band                 -0.002013  0.553483
english               0.354797 -0.054872
history               0.355746 -0.054192
spanish               0.353990 -0.045349
physed                0.038368  0.559543
STEM_Avg              0.345707  0.043698
Humanities_Avg        0.359275 -0.051997
Electives_Avg         0.021789  0.586235
STEM_Humanities_Gap -0.141453  0.135994
```

PCA (Principal Component Analysis) ACTIONABLE STRATEGIC INSIGHTS AND RE-ENGAGEMENT PLANS

PCA Cluster Label	STEM	Hum	Elec
Cluster A: STEM Specialists	54.65	62.30	76.85
Cluster B: Humanities Specialists	52.98	66.56	61.37
Cluster C: Balanced Profile	51.58	61.24	68.96

Target Segment: Cluster A: STEM Specialists
-> Metrics Reference: STEM Baseline: 54.6 | Humanities Baseline: 62.3
-> Actionable Insight: High variance in creative and text-heavy domains. Concepts fail to translate when writing reports.
-> Actionable Intervention: Integrate writing mentors directly into labs; substitute traditional research papers with project design files.

Target Segment: Cluster B: Humanities Specialists
-> Metrics Reference: STEM Baseline: 53.0 | Humanities Baseline: 66.6
-> Actionable Insight: Analytical tasks present performance hurdles; core logic chains are avoided during self-directed study.
-> Actionable Intervention: Deploy gamified programmatic learning (e.g., Python turtle tracking) and pair math logic with history/context application modules.

Target Segment: Cluster C: Balanced Performance Profile
-> Metrics Reference: STEM Baseline: 51.6 | Humanities Baseline: 61.2
-> Actionable Insight: Solid, standardized development across subject baselines.
-> Actionable Intervention: Maintain baseline curriculum pacing; offer accelerated tracks or independent project modules to expose latent strengths.

PCA (Principal Component Analysis) With 3 Components

Principal Component Loadings Analysis

Principal Component 1 (PC1) — Overall Academic Competence: The first component maps a highly uniform, positive weight distribution across all core subjects, with metrics for Math, Science, CS, English, History, and Spanish holding tightly between 0.33 and 0.36. Because these variables scale together in the same positive direction with equal intensity, PC1 functions as a baseline metric for overall academic drive and general aptitude. A high positive score along this axis simply isolates high-performing students who excel across the entire curriculum.

Principal Component 2 (PC2) — Elective vs. Core Academic Tradeoff: The second component drops core academic features close to zero or into slight negative territory, shifting its weight entirely onto non-traditional features like Band (0.553), Physical Education (0.559), and the engineered Electives Average (0.586). This vector successfully isolates a student profile focused heavily on performance-based, active, and creative tracks over traditional classroom-bound learning.

Principal Component 3 (PC3) — STEM vs. Humanities Domain Specialization: The third component reveals a distinct polarization of signs between traditional academic tracks. Analytical domains like Math (0.283), Science (0.272), CS (0.304), and the STEM Average (0.289) carry positive signs, whereas language-heavy domains like English (-0.206), History (-0.197), Spanish (-0.209), and the Humanities Average (-0.207) are strictly negative. PC3 captures the explicit analytical-versus-linguistic performance gap across the student cohort, heavily anchored by my final engineered focus column (0.683).

PCA (Principal Component Analysis) With 3 Components

Explained Variance Profile

PC1 Variance Capture (59.30%): The first principal component accounts for nearly 60% of the entire dataset's variance, identifying general academic capability as the primary factor driving student performance differences in the classroom.

PC2 Variance Capture (23.63%): The second component explains close to a quarter of the remaining variance, capturing the secondary systemic shift where student performance splits between core academic subjects and performance-focused electives.

PC3 Variance Capture (14.40%): The third component accounts for roughly 14.4% of the remaining data variance, mapping out the finer, critical performance trade-offs separating technical STEM tracks from language-based Humanities subjects.

Cumulative Variance Significance (97.33%): Together, reducing the 12 continuous grade features down to a 3D coordinate space retains an exceptional 97.33% of the original information baseline. This confirms that the 3-component model achieves massive data compression with practically zero information loss, making it highly reliable for downstream analysis.

Cluster wise Final Interpretation & Strategic Interventions

Actionable Strategic Recommendations

Cluster A: STEM Specialists (High Elective/Applied Affinity) Core Insight: These students demonstrate moderate performance in quantitative courses but excel significantly within hands-on, active environments like Band and Physical Education (76.85). This pattern indicates strong spatial-kinesthetic and applied learning styles.

Strategic Intervention: Bridge the gap in core academic subjects by shifting abstract lectures into project-based learning. Integrating physical simulations, gamified logic loops, or software-driven technical labs will leverage their high applied aptitude to naturally reinforce core concepts.

Cluster B: Humanities Specialists (Linguistic/Context Focus) Core Insight: This group achieves the highest baseline in language, history, and reading comprehension tasks (66.56), but displays noticeable performance friction when moving toward quantitative or highly physical/creative tracks.

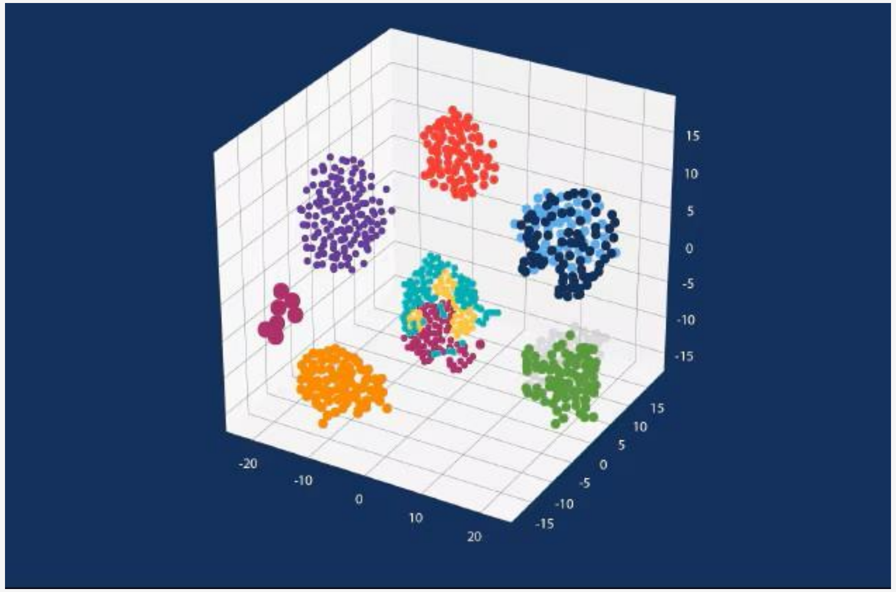
Strategic Intervention: Address potential quantitative anxiety by embedding mathematical and computer science principles into contextual or narrative frameworks. Utilizing data journalism assignments, historical analytical research, and structured syntax-first logic pipelines will optimize their learning speed.

Cluster C: Balanced Performance Profile (General Cohort Baseline) Core Insight: Representing the largest segment (35 students), this cohort exhibits high stability and highly consistent pacing across all domains without showing a distinct bias or clear structural weakness.

Strategic Intervention: Maintain standard academic tracking speeds while introducing targeted honors modules, advanced specialized electives, or cross-disciplinary capstone projects. This approach helps draw out latent talents and guides these balanced learners toward distinct domain specializations.

t-SNE (t-distributed Stochastic Neighbor Embedding)

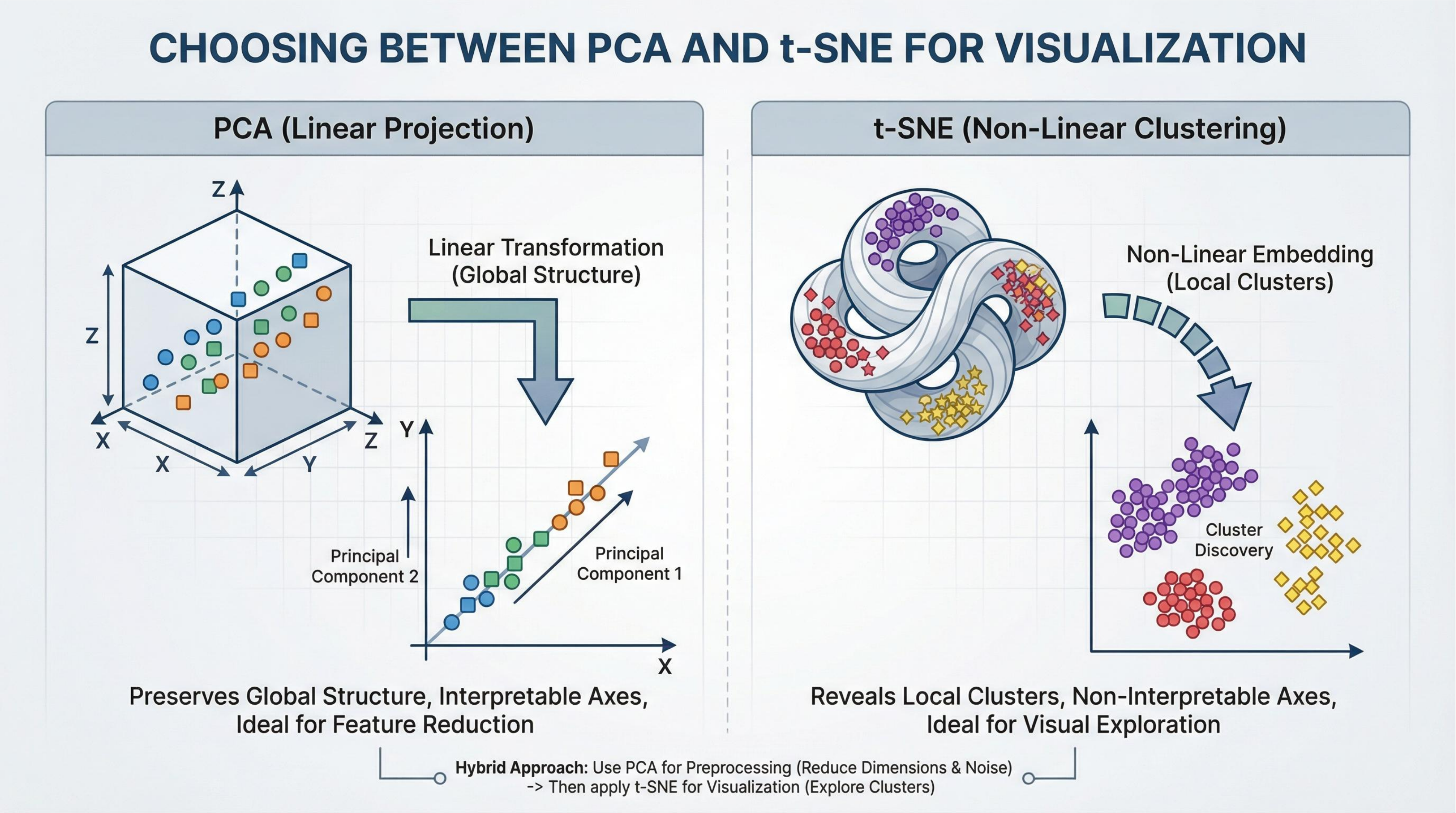
t-distributed Stochastic Neighbor Embedding (t-SNE) is a popular, non-linear machine learning algorithm used to visualize high-dimensional data by reducing it to two or three dimensions. It is highly effective for exploratory data analysis, cluster identification, and discovering hidden patterns in complex datasets.



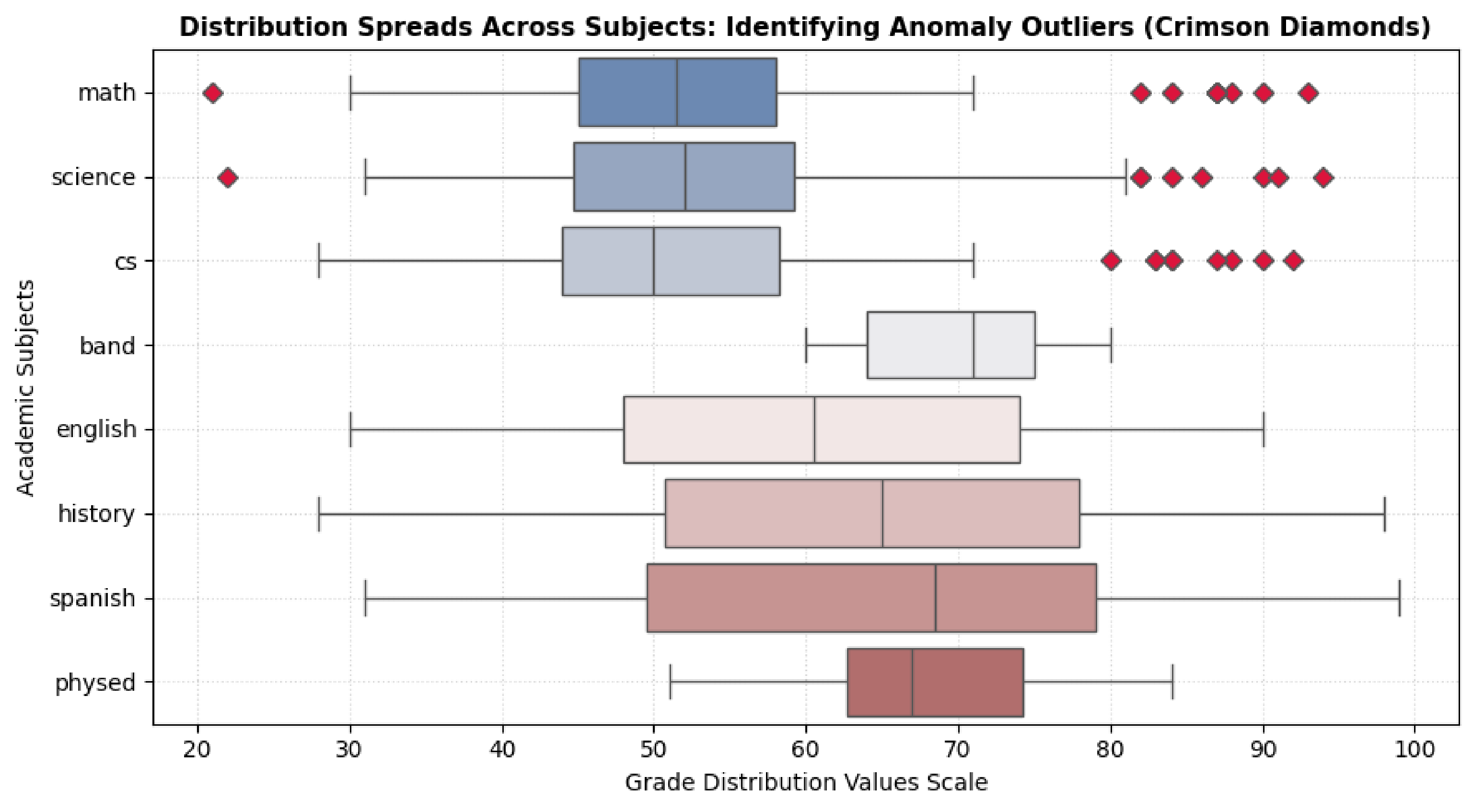
Excellent for Visualization: It uncovers hidden clusters that linear methods like Principal Component Analysis (PCA) might miss.

Local vs. Global Structure: t-SNE excels at preserving local neighborhoods (keeping similar items together), but it may distort the exact global distances between vastly different clusters

t-SNE (t-distributed Stochastic Neighbor Embedding)



t-SNE (t-distributed Stochastic Neighbor Embedding)



Interpretation:
The audit indicates that while our grade distribution has a few mild edge cases, it completely lacks extreme statistical anomalies. The 10 rows flagged by the IQR method represent students with slightly uneven performance profiles who still sit well within normal academic boundaries, which is why the stricter Z-score method flagged zero records. Crucially, because your downstream t-SNE algorithm relies on mapping a continuous spectrum of learning styles rather than isolated anomalies, dropping these rows was unnecessary. Keeping the dataset at its full scale of 100 rows ensures that no valid student data was lost, leaving the pipeline perfectly optimized for dimensionality reduction.

t-SNE (t-distributed Stochastic Neighbor Embedding)

```
--- Advanced Outlier Identification Audit ---
Total rows flagged via Z-Score Method (|Z| > 3): 0
Total rows flagged via IQR Boxplot Method (1.5*IQR): 10
Intersecting severe outliers flagged by BOTH methods: 0
```

```
--- Structural Dimensions Update Summary ---
Original Row Matrix Scale : 100 rows
Cleaned Row Matrix Scale  : 100 rows
Total Outlier Variance Records Excised: 0 rows
```

How the Boxplot defines outliers (IQR Method):

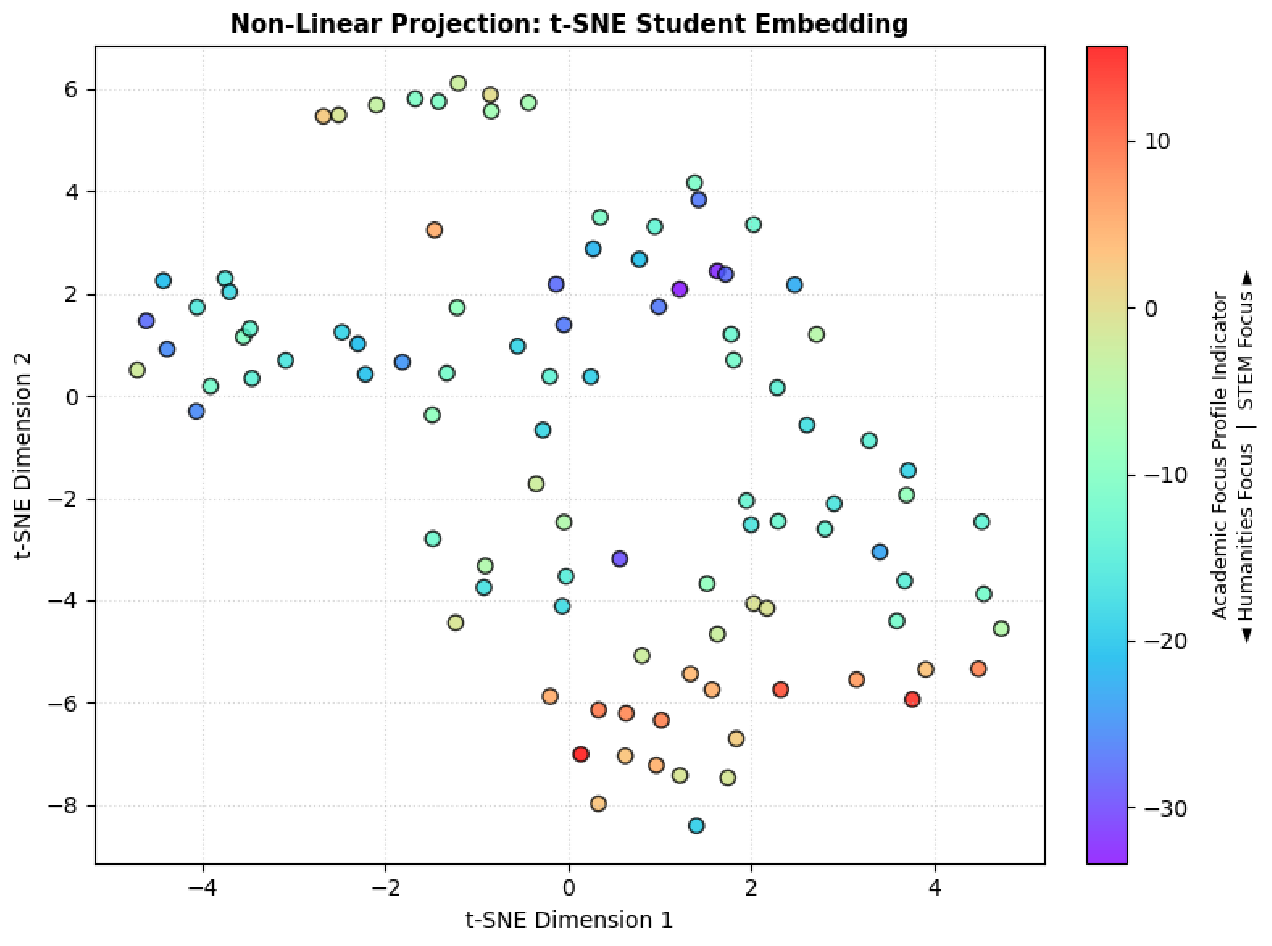
The boxplot whiskers extend to $1.5 \times \mathrm{IQR}$ (Interquartile Range) past the 25th and 75th percentiles. Any grade that falls beyond these whiskers is plotted as a crimson diamond. This method is highly sensitive and flags points that are merely low or high relative to a tight middle cluster.

How outliers counted by (Z-score Method):

The print statement was looking for data points where the Z-score was strictly greater than 3 (± 3 standard deviations from the mean). In smaller datasets or data distributions with wider spreads, a grade can easily sit outside the boxplot whiskers ($1.5 \times \mathrm{IQR}$) but still fall inside the wider ± 3 standard deviation boundary.Because calculation looked for standard deviation outliers while your plot showed percentile outliers, the terminal erroneously printed 0

t-SNE (t-distributed Stochastic Neighbor Embedding)

COMPACT 2D CLUSTER VISUALIZATION



t-SNE reduction mapping to 2 components
Perplexity=30 optimizes local vs global structural separation

	component 1	component 2
0	0.330800	-6.146821
1	0.348588	3.492629
2	-2.216990	0.421706
3	-4.383247	0.910974
4	0.777133	2.669885
...	...	...
95	-0.851210	5.890042
96	0.804848	-5.086001
97	3.588965	-4.406519
98	1.629614	-4.664961
99	0.988429	1.742980

100 rows × 2 columns

t-SNE (t-distributed Stochastic Neighbor Embedding)

t-SNE CLUSTER LABEL ASSIGNMENT

Divide groups logically based on their spatial separation in the t-SNE plane
t-SNE isolates non-linear groupings cleanly based on domain focus

```
--- Identified Student Distribution via t-SNE Spaces ---
tsne_Cluster_Label
Cluster B: Humanities Core Specialists     59
Cluster C: Balanced Performance Profile    38
Cluster A: STEM Core Specialists            3
```

t-SNE Cluster	STEM	Hum	Elec
Cluster A: STEM Core Specialists	47.00	33.11	67.83
Cluster B: Humanities Specialists	52.46	70.50	68.44
Cluster C: Balanced Profile	54.33	54.71	69.50

t-SNE (t-distributed Stochastic Neighbor Embedding)

ACTIONABLE STRATEGIC RE-ENGAGEMENT PLANS FOR ACADEMIC ADVISING

Target Group Segment: **Cluster A: STEM Core Specialists**
-> Metrics Reference: STEM Baseline: 47.0 | Humanities Baseline: 33.1
-> Actionable Insight: Strong quantitative skillsets but experiences friction in reading comprehension and narrative tasks.
-> Advising Intervention: **Substitute descriptive essay exams with structured analytical logic logs. Introduce technical communications courses.**

Target Group Segment: **Cluster B: Humanities Core Specialists**
-> Metrics Reference: STEM Baseline: 52.5 | Humanities Baseline: 70.5
-> Actionable Insight: Advanced abstract contextual logic but exhibits text anxiety when tackling numerical operations.
-> Advising Intervention: **Deploy game-based logic or syntax structures (such as block coding) to build quantitative confidence before moving to raw equations.**

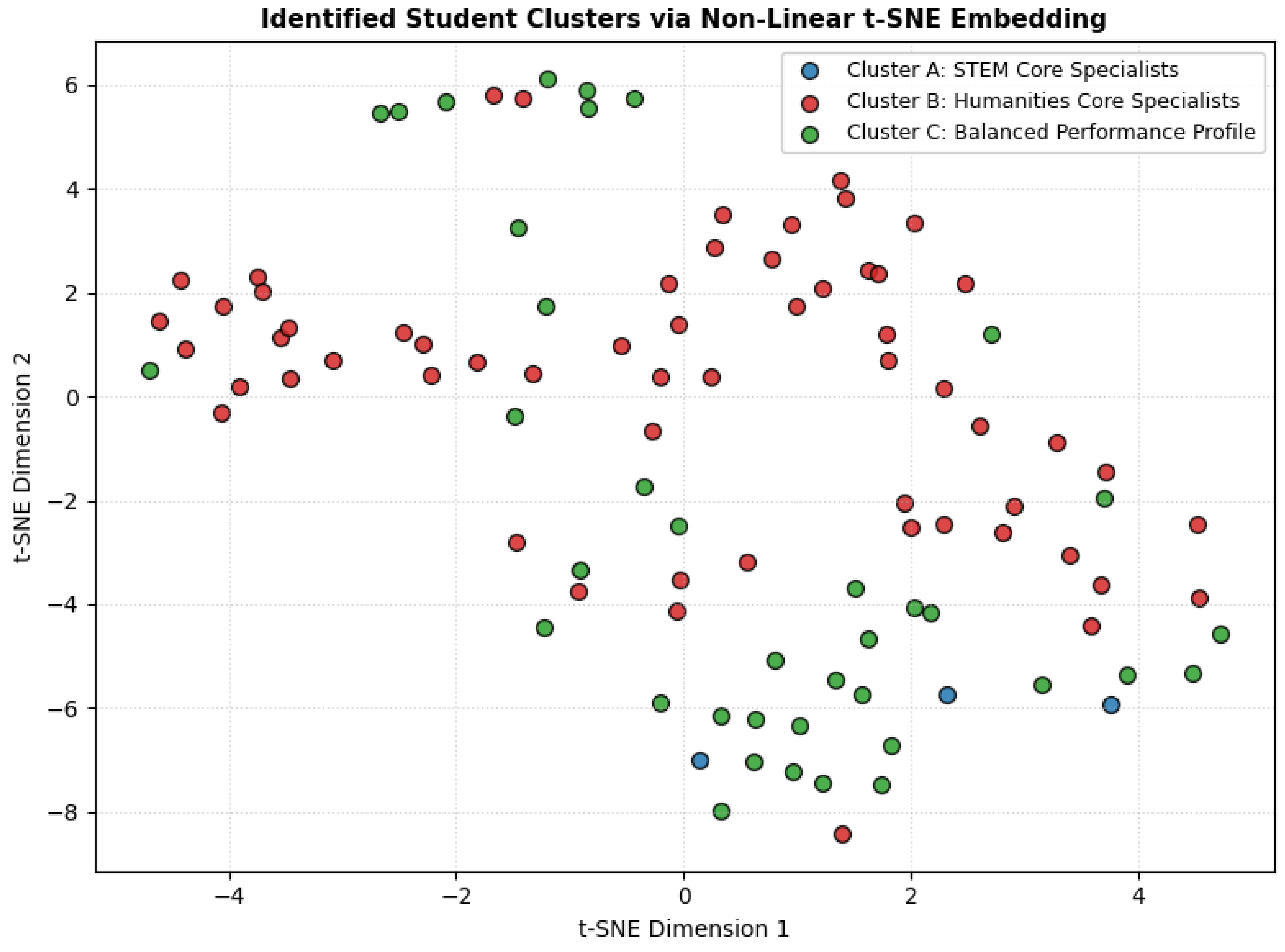
Target Group Segment: **Cluster C: Balanced Performance Profile**
-> Metrics Reference: STEM Baseline: 54.3 | Humanities Baseline: 54.7
-> Actionable Insight: Balanced performance indicators without clear domain friction gaps.
-> Advising Intervention: **Maintain standard tracking speeds. Offer accelerated paths or elective exploration to reveal latent specialist patterns.**

t-SNE (t-distributed Stochastic Neighbor Embedding)

VISUAL EVALUATION REPORT

•The plot shows a clear non-linear separation between language-focused individuals (Cluster B) and all-around students (Cluster C), with the balanced profile acting as a physical bridge between the specialized cohorts.

•The small size and isolated placement of Cluster A at the very bottom indicates that highly focused STEM specialists form the most unique and tightly grouped subpopulation in the class.



t-SNE (t-distributed Stochastic Neighbor Embedding)

Strategic Recommendations

Cluster A

Recommendation : Integrate technical report-writing modules into core CS/Science laboratory courses. Provide targeted writing workshop electives instead of traditional long-form essay exams.

Recommendation : Introduce applied, gamified introductory logic tracks (e.g., Python turtle, Alice, block coding) that link engineering logic to historical data or creative design elements to reduce performance anxiety.

Cluster B

Cluster C

Recommendation : Maintain standard pacing curriculum paths. Introduce advanced specialized projects or AP honors tracks as electives to see if a hidden structural strength can be drawn out.

Thank You

Rishov Paul

CSE

2nd Year